STA 431 Midterm II

FRONT: DISTRIBUTIONS + MOMENTS

Handwritten-copy master • US Letter, landscape • Three columns on each side

1 Joint, marginal, conditional

Normalize first. PMF: $p(x,y) \geq 0$, $\sum_x \sum_y p(x,y) = 1$. PDF: $f(x,y) \geq 0$, $\iint_S f = 1$. If $f = cq$, then $c = 1/\iint_S q$ (use sums for PMFs). Write the support S; zero elsewhere.

Marginal = eliminate the other variable.

$$p_X(x) = \sum_y p(x,y), \qquad f_X(x) = \int_{y:(x,y)\in S} f(x,y)\,dy.$$

Interchange x, y for the other marginal. A marginal's support is the projection of S onto its axis.

Conditional = joint / conditioning marginal.

$$p_{X|Y}(x \mid y) = \frac{p(x,y)}{p_Y(y)}, \qquad f_{X|Y}(x \mid y) = \frac{f(x,y)}{f_Y(y)}.$$

Use only where the denominator is positive; retain the slice of S at that y. Sum/integrate to 1 over x.

$$f(x,y) = f_{X|Y}(x \mid y) f_Y(y).$$

A conditional alone or two marginals generally do *not* determine the joint. Conditional plus its conditioning marginal does (almost everywhere).

Joint CDF. $F(x,y) = P(X \leq x, Y \leq y)$.

$$F(x,y) = \sum_{u\leq x} \sum_{v\leq y} p(u,v)$$

$$F(x,y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u,v)\,dv\,du.$$

Intersect this rectangle with S before integrating.

$$F_X(x) = \lim_{y\to\infty} F(x,y), \quad F_Y(y) = \lim_{x\to\infty} F(x,y).$$

For an absolutely continuous joint law, $f(x,y) = \partial^2 F/\partial x \partial y$ almost everywhere. For integer-valued variables,

$$p(x,y) = F(x,y) - F(x-1,y) - F(x,y-1) + F(x-1,y-1).$$

Rectangle from any CDF:

$$P(a < X \leq b, c < Y \leq d) = F(b,d) - F(a,d) - F(b,c) + F(a,c).$$

For discrete inclusive lower endpoints use left limits (or $a-1, c-1$ on an integer lattice). Continuous endpoints have zero mass.

2 Events, bounds, independence

Region method. Translate words → event A; draw $S \cap A$; select order; find outer range; then inner slice limits.

$$P((X,Y) \in A) = \sum_{(x,y)\in S\cap A} p(x,y) = \iint_{S\cap A} f(x,y)\,dx\,dy.$$

An empty slice contributes zero; split where a boundary changes. The inner limits may depend on the outer variable; the outer limits cannot depend on the inner variable.

Common regions. On $0 < x < 1, 0 < y < 2$:

$$P(X < Y) = \int_0^1 \int_x^2 f(x,y)\,dy\,dx.$$

Equivalently split at $y = 1$:

$$\int_0^1 \int_0^y f\,dx\,dy + \int_1^2 \int_0^1 f\,dx\,dy.$$

On $x > 0, y > 0, x + y < 1$:

$$\iint_S f = \int_0^1 \int_0^{1-x} f\,dy\,dx = \int_0^1 \int_0^{1-y} f\,dx\,dy.$$

For $P(X + Y \leq a)$, intersect $y \leq a - x$ with the support. For $P(X > Y)$, use $y < x$. In a PMF list actual allowed pairs; do not integrate.

Words → notation.

- Both lifetimes exceed t: $X > t, Y > t$.
- Series device fails if either fails: $T = \min(X,Y)$; $P(T \leq t) = 1 - P(X > t, Y > t)$.
- Parallel device fails only if both fail: $T = \max(X,Y)$; $P(T \leq t) = P(X \leq t, Y \leq t)$.
- At least a given $Y = y$: integrate $f_{X|Y}$ over $x \geq a$ in its slice.

For an event B of positive probability, $P(A \mid B) = P(A \cap B)/P(B)$. For continuous conditioning $Y = y$, use the conditional density; $P(Y = y) = 0$.

Independence. For all x, y,

$$F(x,y) = F_X(x) F_Y(y).$$

Equivalent: $p(x,y) = p_X(x)p_Y(y)$ at every pair; or $f(x,y) = f_X(x)f_Y(y)$ almost everywhere, *including zero regions*. One failing pair (PMF) or a positive-area failure (PDF) disproves it. Factorable formula on triangular support is insufficient. Independence makes the conditional equal its marginal and permits factorization of event probabilities and expectations of products.

3 Expectation, moments, MGFs

Expectation / LOTUS.

$$\mathrm{E}[g(X)] = \sum_x g(x) p_X(x) \quad \text{or} \quad \int g(x) f_X(x)\,dx.$$

For $g(X,Y)$ use $\sum\sum gp$ or $\iint_S gf$. Finite expectation requires $\mathrm{E}\,|g(X)| < \infty$.

$$\mathrm{E}[aX + bY + c] = a\,\mathrm{E}\,X + b\,\mathrm{E}\,Y + c.$$

Linearity needs no independence. In general $\mathrm{E}[g(X)] \neq g(\mathrm{E}\,X)$. For an indicator, $\mathrm{E}[1_A] = P(A)$. If $g \geq 0$, then $\mathrm{E}\,g \geq 0$.

Raw and central moments.

$$\mu_r' = \mathrm{E}[X^r], \quad \mu = \mu_1', \quad \mu_r = \mathrm{E}[(X-\mu)^r].$$

$$\mu_0' = 1, \quad \mu_1 = 0, \quad \mathrm{Var}\,X = \mu_2 = \mu_2' - \mu^2.$$

$$\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}\,X, \quad \sigma_{aX+b} = |a|\sigma_X.$$

For $Y = g(X)$, compute $\mathrm{E}\,g$ and $\mathrm{E}\,g^2$; then $\mathrm{Var}\,Y = \mathrm{E}[g(X)^2] - (\mathrm{E}\,g(X))^2$. For $Y = X^2$, variance requires $\mathrm{E}\,X^4$. If density/support are symmetric about zero, existing odd moments vanish; even moments need integration.

MGF. $M_X(t) = \mathrm{E}[e^{tX}]$.

$$M_X(t) = \sum_x e^{tx} p_X(x) \quad \text{or} \quad \int e^{tx} f_X(x)\,dx.$$

State the set of t where it is finite. The usual moment and uniqueness results require finiteness on an open interval containing 0.

$$M(0) = 1, \quad M^{(r)}(0) = \mathrm{E}\,X^r, \quad \mathrm{Var}\,X = M''(0) - [M'(0)]^2.$$

Equal MGFs on a neighborhood of zero imply equal distributions. All moments existing does not guarantee such an MGF.

$$M_{aX+b}(t) = e^{bt} M_X(at).$$

For independent X_i,

$$M_{\sum a_i X_i + b}(t) = e^{bt} \prod_i M_{X_i}(a_i t).$$

Intersect the transformed domains. Without independence, do not multiply marginal MGFs.

Fast templates.

Law/form	$M(t)$	$\mathrm{E}\,X,\ \mathrm{Var}\,X$
$\lambda e^{-\lambda x},\ x > 0$	$\frac{\lambda}{\lambda - t},\ t < \lambda$	$1/\lambda,\ 1/\lambda^2$
$e^{\alpha t + \beta t^2}$	all t	$\alpha,\ 2\beta$

$\int_0^\infty x^n e^{-ax}\,dx = n!/a^{n+1}$ for $a > 0$, integer $n \geq 0$.

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BACK: COVARIANCE + CONDITIONING + TI-84

Use the formulas to set up exact work; use numerical commands to check arithmetic and one-dimensional integrals.

4 Product moments + linear combinations

Product moments.

$$\mu'_{rs} = \mathrm{E}[X^r Y^s], \qquad \mu_{rs} = \mathrm{E}[(X-\mu_X)^r (Y-\mu_Y)^s].$$

Compute from the joint law, not by multiplying marginals unless independence is justified.

$$\mathrm{Cov}(X,Y) = \mathrm{E}[XY] - \mu_X \mu_Y, \qquad \rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}.$$

Correlation needs positive finite variances; $-1 \le \rho \le 1$. $|\mathrm{Cov}(X,Y)| \le \sigma_X \sigma_Y$.

$$\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X), \quad \mathrm{Cov}(X,X) = \mathrm{Var}\, X,$$
$$\mathrm{Cov}(aX+b, cY+d) = ac\,\mathrm{Cov}(X,Y).$$

Independent $\Rightarrow$ zero covariance (finite second moments). Zero covariance $\not\Rightarrow$ independence. Example: X symmetric on $(-1,1)$, $Y = X^2$; $\mathrm{E}\, X = \mathrm{E}\, X^3 = 0$, hence $\mathrm{Cov}(X,Y) = 0$, yet Y is determined by X. To prove dependence use events A, B for which $P(A \cap B) \ne P(A)P(B)$.

One combination. $W = c + \sum_i a_i X_i$:

$$\mathrm{E}\, W = c + \sum_i a_i \mu_i,$$
$$\mathrm{Var}\, W = \sum_i a_i^2 \sigma_i^2 + 2 \sum_{i<j} a_i a_j \sigma_{ij}.$$

Here $\sigma_{ij} = \mathrm{Cov}(X_i, X_j)$. Write each unordered covariance pair once, times 2.

$$\mathrm{Var}(aX+bY+cZ) = a^2\,\mathrm{Var}\, X + b^2\,\mathrm{Var}\, Y + c^2\,\mathrm{Var}\, Z + 2ab\,\mathrm{Cov}(X,Y) + 2ac\,\mathrm{Cov}(X,Z) + 2bc\,\mathrm{Cov}(Y,Z).$$

Negative coefficients change cross-term signs; constants contribute no variance. Uncorrelated variables suffice to remove the cross terms. Independence is stronger.

Two combinations. $U = aX + bY + c$, $V = dX + eY + f$:

$$\mathrm{Cov}(U,V) = ad\,\mathrm{Var}\, X + be\,\mathrm{Var}\, Y + (ae+bd)\,\mathrm{Cov}(X,Y).$$

More generally $\mathrm{Cov}(\sum_i a_i X_i, \sum_j b_j X_j) = \sum_i \sum_j a_i b_j \sigma_{ij}$. Find ρ_{UV} by dividing this covariance by $\sqrt{\mathrm{Var}\, U\,\mathrm{Var}\, V}$.

5 Conditional moments + workflows

Conditional expectation. Fix the conditioned value, then average over the variable that remains random:

$$m(y) = \mathrm{E}[X \mid Y = y] = \sum_x x\, p_{X|Y}(x \mid y) \quad \text{or} \quad \int x f_{X|Y}(x \mid y)\, dx.$$

For $\mathrm{E}[g(X) \mid Y = y]$, replace x by $g(x)$.

$$q(y) = \mathrm{E}[X^2 \mid Y = y],$$
$$\mathrm{Var}(X \mid Y = y) = q(y) - [m(y)]^2.$$

Do not square before taking the conditional second moment. $\mathrm{E}[X \mid Y]$ is a random variable $m(Y)$; $m(y_0)$ is a number.

Marginal-first recipe.

1. Find $f_Y(y)$ or $p_Y(y)$ with its support.
2. Divide joint by it, retaining the conditional slice.
3. Integrate/sum x and x^2 against that law.
4. Evaluate at the specified y; subtract $m(y)^2$ for variance.

Direct ratio shortcut.

$$\mathrm{E}[g(X) \mid Y = y] = \frac{\int g(x) f(x,y)\, dx}{f_Y(y)}$$

(use sums for discrete). Denominator must be positive. Reconstruct marginal moments by averaging over Y:

$$\mathrm{E}\, X = \mathrm{E}[\mathrm{E}(X \mid Y)],$$
$$\mathrm{Var}\, X = \mathrm{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}[\mathrm{E}(X \mid Y)].$$

Independence gives $\mathrm{E}[g(X) \mid Y] = \mathrm{E}\, g(X)$ when integrable. A constant conditional mean alone does not prove independence.

Problem-solving order.

1. Identify PMF / PDF / CDF / MGF / supplied moments.
2. Record support, event, and conditioning direction.
3. Normalize if a constant is unknown.
4. Select formula; show the sum/integral and bounds.
5. Keep fractions through the algebra; check the result.

Checks. Probabilities in $[0,1]$; PMFs sum to 1; PDFs integrate to 1 (a PDF value may exceed 1); variances nonnegative; $\mathrm{E}\, X$ inside the range of X when finite; $M(0) = 1$; correlation in $[-1,1]$.

Useful algebra. $\sum_{k=1}^n k = n(n+1)/2$; $\sum_{k=1}^n k^2 = n(n+1)(2n+1)/6$; $\sum_{k=1}^n k^3 = [n(n+1)/2]^2$.

6 TI-84 Plus CE: useful checks

Find commands. `[2nd][0]` opens CATALOG. Select the actual command token. `MATH`: `nDeriv(`, `fnInt(`; `[2nd][STAT]` LIST: `OPS` → `seq(`, `MATH` → `sum(`. Use `MODE: CLASSIC` for the linear syntax below; MathPrint instead shows templates.

Finite PMF: values in L_1, probabilities in L_2.

`sum(L2)` (must be 1)

`sum(L1*L2)` ($\mathrm{E}\, X$)

`sum(L1^2*L2)-sum(L1*L2)^2` ($\mathrm{Var}\, X$)

For unnormalized nonnegative weights w in L_2, divide each moment sum by `sum(L2)` first.

Generate finite sums without lists.

`sum(seq(expression,K,first,last))`

HW5-style $p(x) = (x^2+x)/40$, $x = 1, \ldots, 4$:

`sum(seq(K*(K^2+K)/40,K,1,4))`

gives $\mathrm{E}\, X = 13/4$. For $\mathrm{E}\, X^2$, use K^2 instead of the leading K. For $\mathrm{Var}(X^2)$, obtain $\mathrm{E}\, X^4 - (\mathrm{E}\, X^2)^2$.

Joint PMF as rows of allowed pairs. Put all x values in L_1, corresponding y in L_2, and joint probabilities in L_3.

`sum(L1*L2*L3)` ($\mathrm{E}\, XY$)

`sum((L1>L2)*L3)` ($P(X > Y)$)

`sum((L1+L2<=3)*L3)` ($P(X+Y \le 3)$)

Use TEST's single $\le$ token for `<=`; tokens: `[2nd][MATH]` TEST. Every joint pair needs its own row; do not pair just the distinct marginal values.

One-dimensional numerical integrals.

`fnInt(expression,X,lower,upper)`

HW5 check: `fnInt(X^2*3/4*(1-X^2),X,-1,1)`

gives $1/5$. Replacing X^2 by X^4 gives $3/35$. For $f(x,y) = x + y$ on the unit square, at $y = 1/2$, conditional density is $x + 1/2$:

`fnInt(X*(X+1/2),X,0,1)` ($7/12$)

`fnInt(X^2*(X+1/2),X,0,1)` ($5/12$)

MGF first-derivative check.

`nDeriv(3/(3-X),X,0)` ($\approx 1/3$)

X here is the MGF's argument t. Derive $M''(0)$ by hand; avoid nested numerical derivatives.

Limits. No native symbolic algebra. Do the inner integral by hand before `fnInt`; numerical output does not prove independence or MGF existence. `1-Var Stats` with weights uses σ_x^2 for population variance; `Sx^2` and `variance(` are sample quantities. Prefer weighted sums above. `MATH` → `▶Frac` may suggest a fraction; it does not prove exactness.